



IF3170 Inteligensi Artifisial

Institut Teknologi Bandung

Introduction to Learning and Supervised Learning

Semester I TA 2026/2027





Outline

- Review:
 - What is AI
 - Intelligent Agent: PEAS, Property of Task Environment
 - Local Search
 - Adversarial Search
 - Constraints Satisfaction Problem (CSP)
 - Introduction to Knowledge Based Systems (KBS)
 - Rule Based Systems (RBS)
- Introduction to Learning
- Supervised Learning





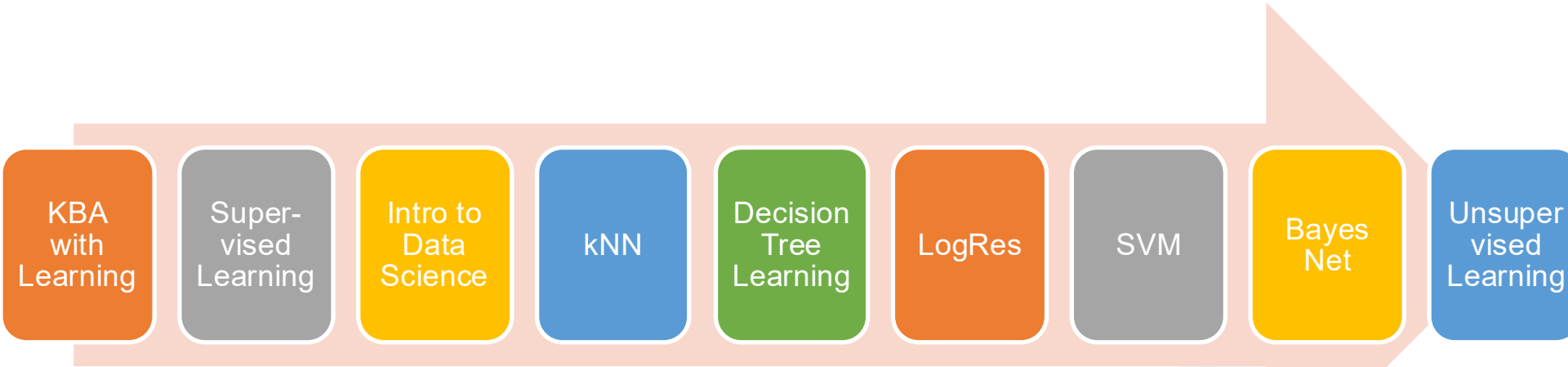
Problem Solving Agent & Knowledge Based Agent

Percepts used to get solution (acting)

- Problem Solving Agent
 - Design: formulate problem → search solution → execute
 - Agents know world dynamics, states are finite, deterministic, optimal state value
- Knowledge-based Agent
 - Agents know world dynamics, states are infinite, deterministic
 - Design: knowledge base + inference engine
 - Manual knowledge acquisition (for knowledge base): elicitation + representation



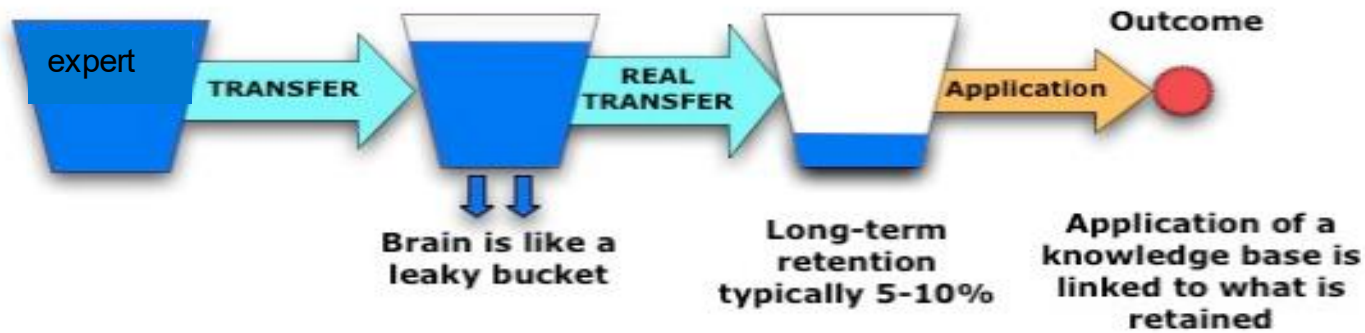
Learning





Knowledge Acquisition Bottleneck

- Knowledge elicitation:
 - slow speed
 - Inability of expert to express the knowledge they possess
- Knowledge representation: **knowledge mismatch**
 - Perbedaan struktur pengetahuan pakar dan struktur pengetahuan program



<http://www.profoundlearning.com/Content/EducationSolutions/educationKnowledgeTransfer.html>



Knowledge-based Agent with Learning

Starting with an empty knowledge-base

Agent designer can TELL sentences one by one

agent knows how to operate in its environment

Learning allows agent to operate in initially unknown environments

Learning improves its performance on future tasks after making observation about the world



Why Need Learning?

Learning is essential for unknown environments,

- i.e., when designer lacks omniscience, agent doesn't know world dynamics

Learning is useful as a system construction method,

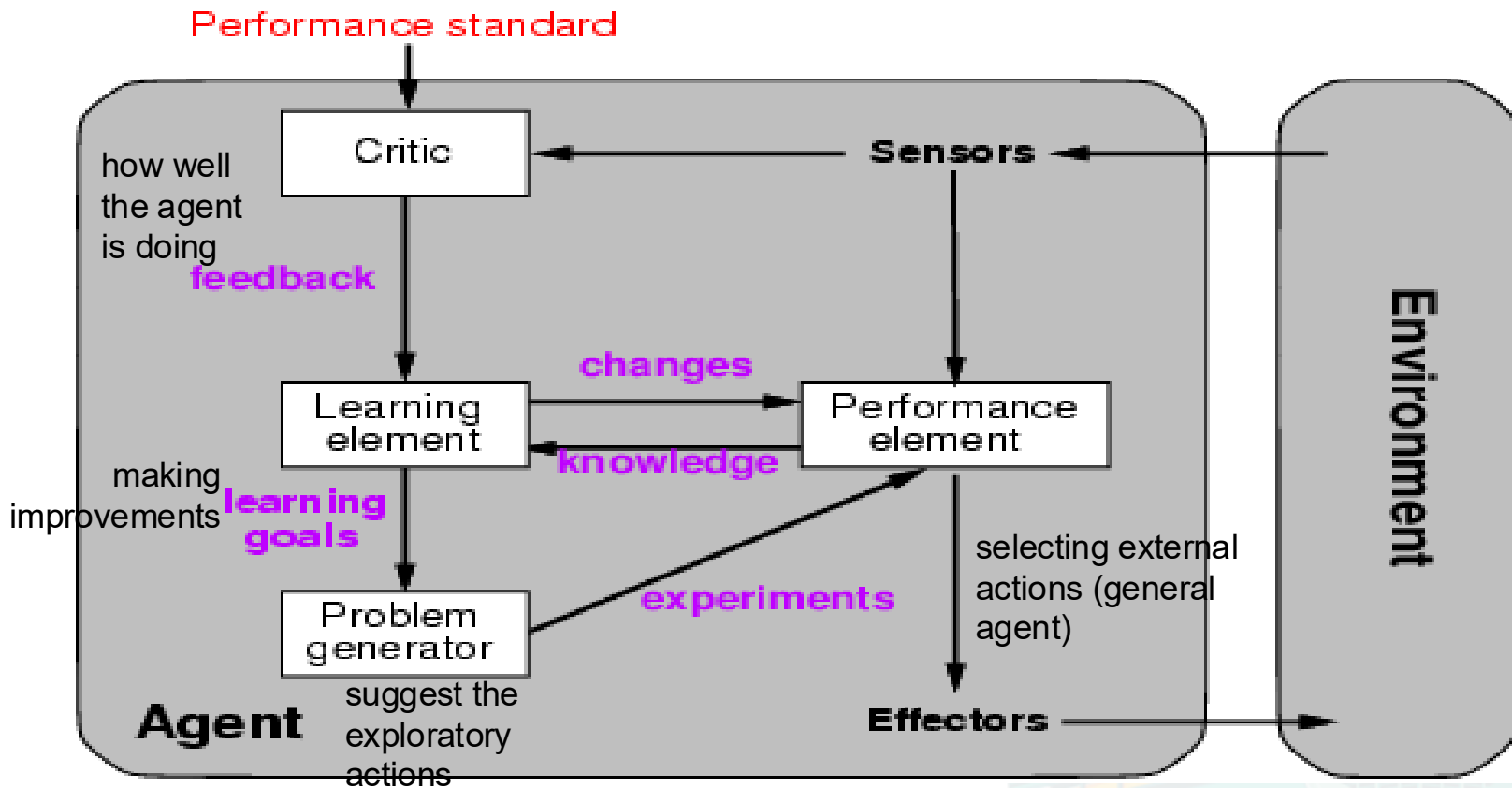
- i.e., expose the agent to reality rather than trying to write it down

Learning modifies the agent's decision mechanisms to improve performance

- Learning from observations, Feedback for improving the agent's ability to act in the future



KBA with Learning = Learning Agent



Russell, S. J., & Norvig, P. (2010). Artificial Intelligence-A Modern Approach, Third International Edition.



Design of A Learning Element

Learning: **changes** in the system that are **adaptive** in the sense that they enable the system to do the task or tasks drawn from the same population **more efficiently** and **more effectively** the next time.

Which component
of performance
element are to be
learned

Availability of prior
knowledge

What feedback is
available to learn
these components

What
representation is
used for the
components



Taxi Driver Agent: PE & Feedback



<http://www.gettyimages.com/detail/83988175/Stone>



<http://www.stahle.com/>

- Brake decision:
 - f: states $\rightarrow$ boolean (brake or not)
 - Feedback: instructor shouts
- Buses recognition:
 - f: images $\rightarrow$ boolean
 - Feedback: labeling bus images
- Good/bad traffic day recognition
 - Develop own concepts
 - No feedback
- Desirable/undesirable behavior recognition from tip indication
 - Feedback: tip from customer at the end of journey



Learning Type based on Feedback

Unsupervised Learning (no feedback)

- Given set of examples without label, detect potentially useful clusters of input examples. E.g. good/bad traffic day recognition.

Supervised Learning

- Given set of examples (input-output pairs), learns a function that maps from input to output. E.g. brake decision

Reinforcement Learning

- Agent learns from a series of reinforcements (rewards or punishments). It is up to agent to decide which if the actions prior to the reinforcement were most responsible for it.



Summary

KBA with
learning =
learning agent

Why: unknown
environment, system
construction method,
improve performance

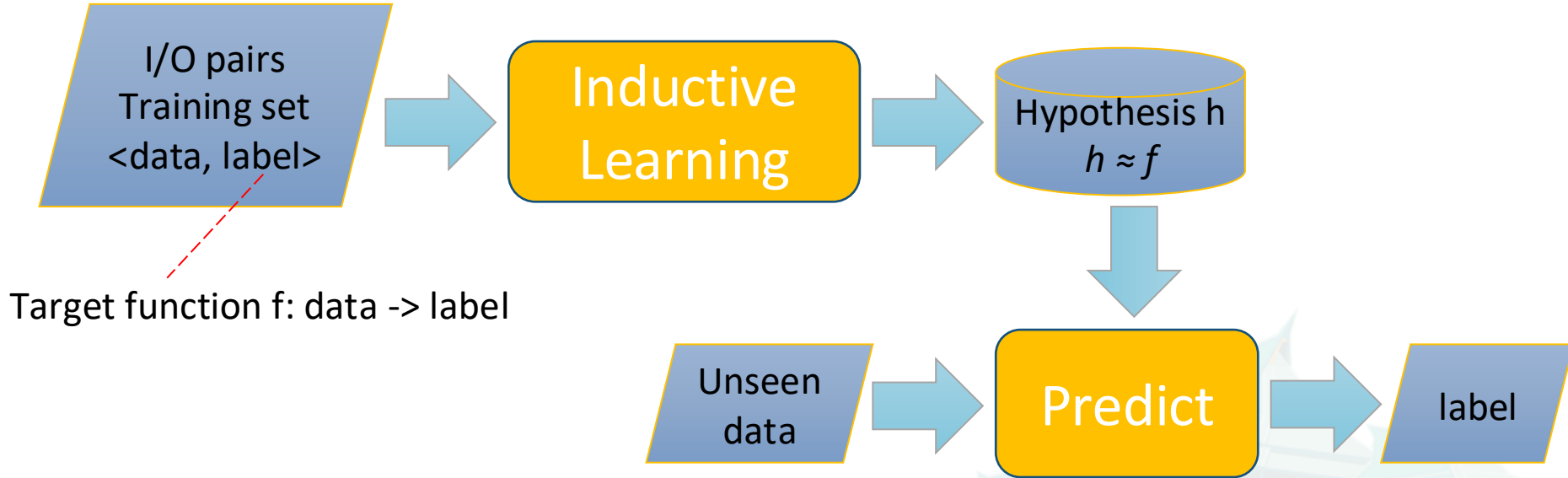
Design learning
element: which
PE, prior knowl,
feedback,
representation

Learning type:
supervised,
unsupervised,
reinforcement

Supervised Learning



Supervised Learning





Supervised Learning (2)

- Learn a function from training data
- Input: training-set $\langle \text{data}, \text{label} \rangle \rightarrow \langle x, y \rangle$
- Function of target learning: $f: x \rightarrow y$
- Problem: find a **hypothesis** h s.t $h \approx f$
- Learning a (possibly incorrect) general function or rule from specific input-output pairs is called inductive learning
 - Not analytical or deductive learning





Supervised Learning (3)

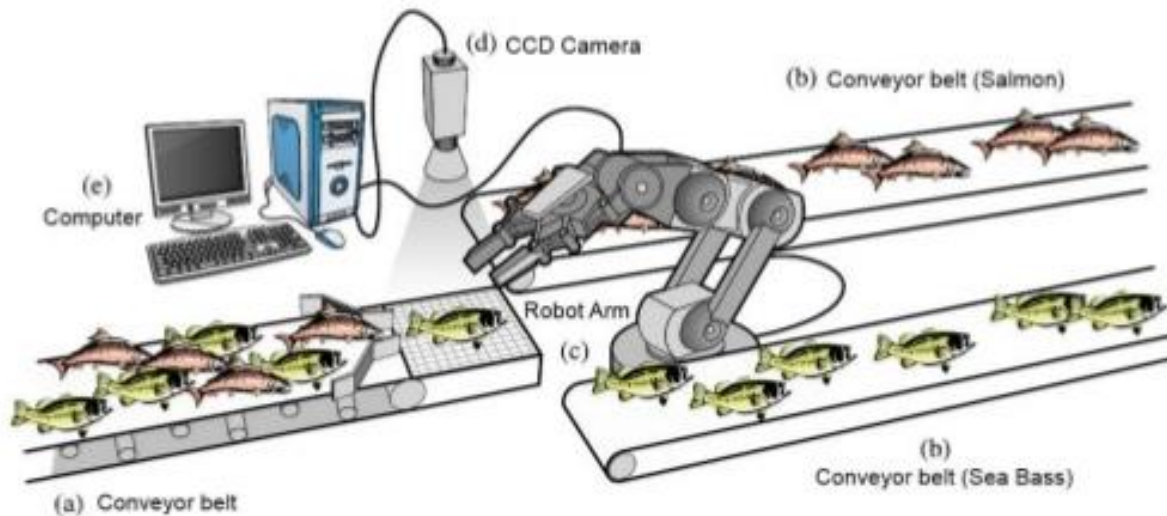
- Process: inductive learning
- Learning output: h (class description)
 - $f: x \rightarrow y$
 - Function f can be stochastic: $P(Y|x)$
- Learning problem:
 - Classification: domain (y) is a finite set of values (boolean, classes)
 - Regression: domain (y) is number (temperature, stock prices)
- Usage: classify unseen data based on class description



Fish Packing Plant

Fish packing plant wants to automate the process of sorting incoming fish on a conveyor belt according to species.

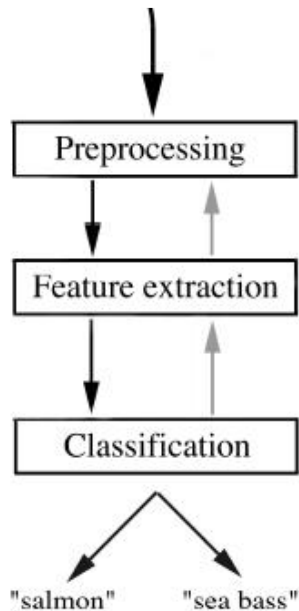
- A: Conveyor belt for fish
- B: Conveyor belt for classified fish
- C: Robot arm for grabbing fish
- D: Machine vision system with CCD camera
- E: Computer that analyze fish image and control the robot arm



Automated Fish Classification System



Fish Classification Agent



- Comp. of PE: fish classification
- Feedback: labeling fish
- Klasifikasi: model klasifikasi
 - Hasil supervised learning



Dataset Construction (Preprocessing)



Enhancement

Segmentasi

Resize



Duda dkk, 2001



Fish features

- Piksel
- Descriptor based on : color, shape, textures

Find physical differences between the two types of fish:

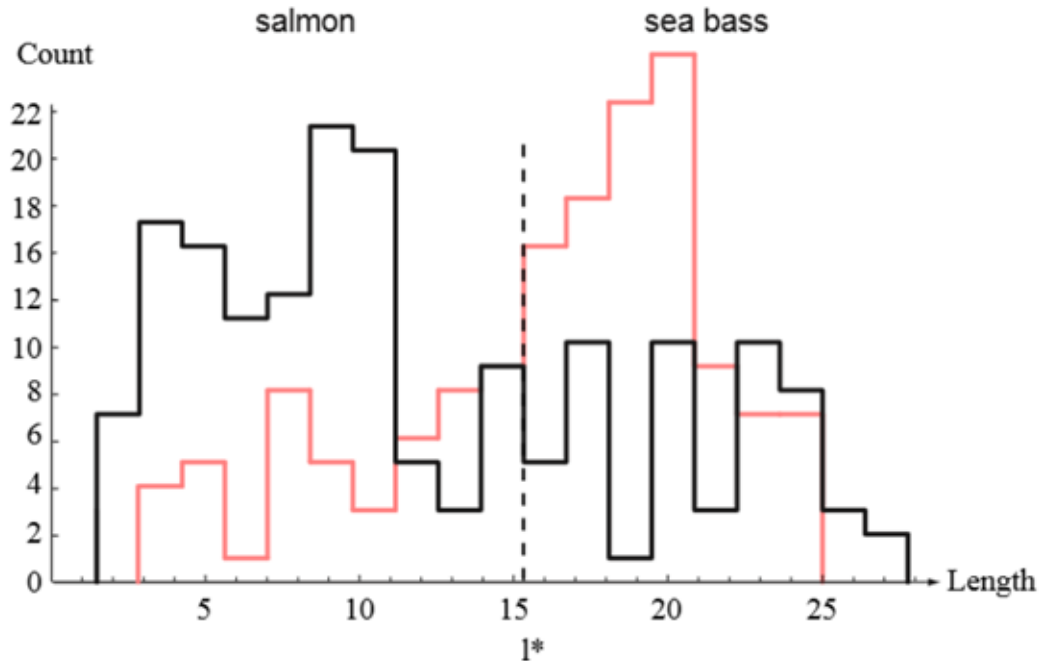
- Length
- Lightness
- Width
- Number and shape of fins
- Position of the mouth,
- ...





Tentative Model: Length Feature

Suppose somebody at the fish plant tells us that a sea bass is generally longer than a salmon.



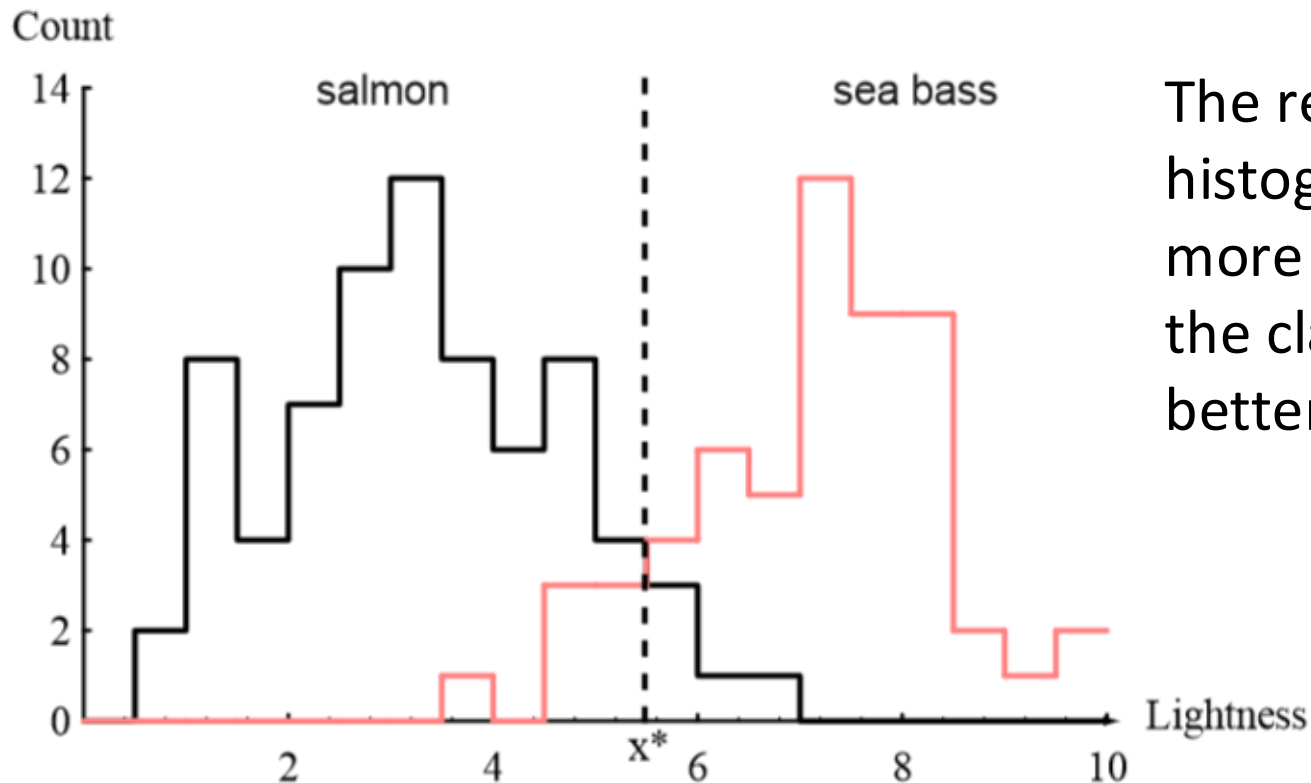
From histograms, sea bass are longer than salmon, on average, but it is clear that this single criterion is quite poor;

No matter how we choose l^* , we cannot reliably separate sea bass from salmon by length alone.

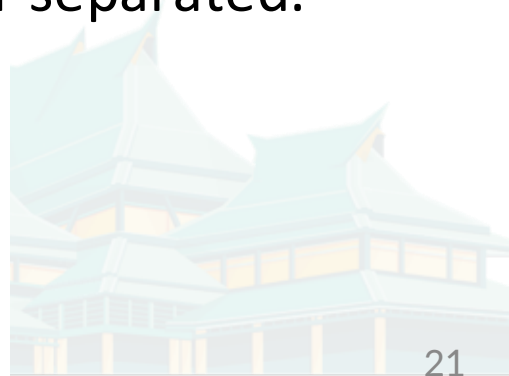
The value l^* marked will lead to the smallest number of errors, on average.



Tentative Model: Lightness Feature

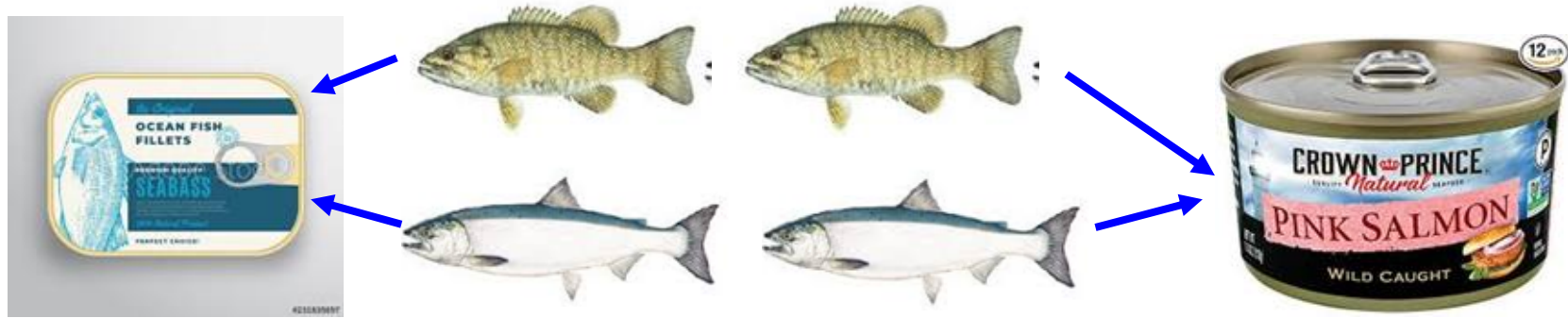


The resulting histograms are much more satisfactory — the classes are much better separated.





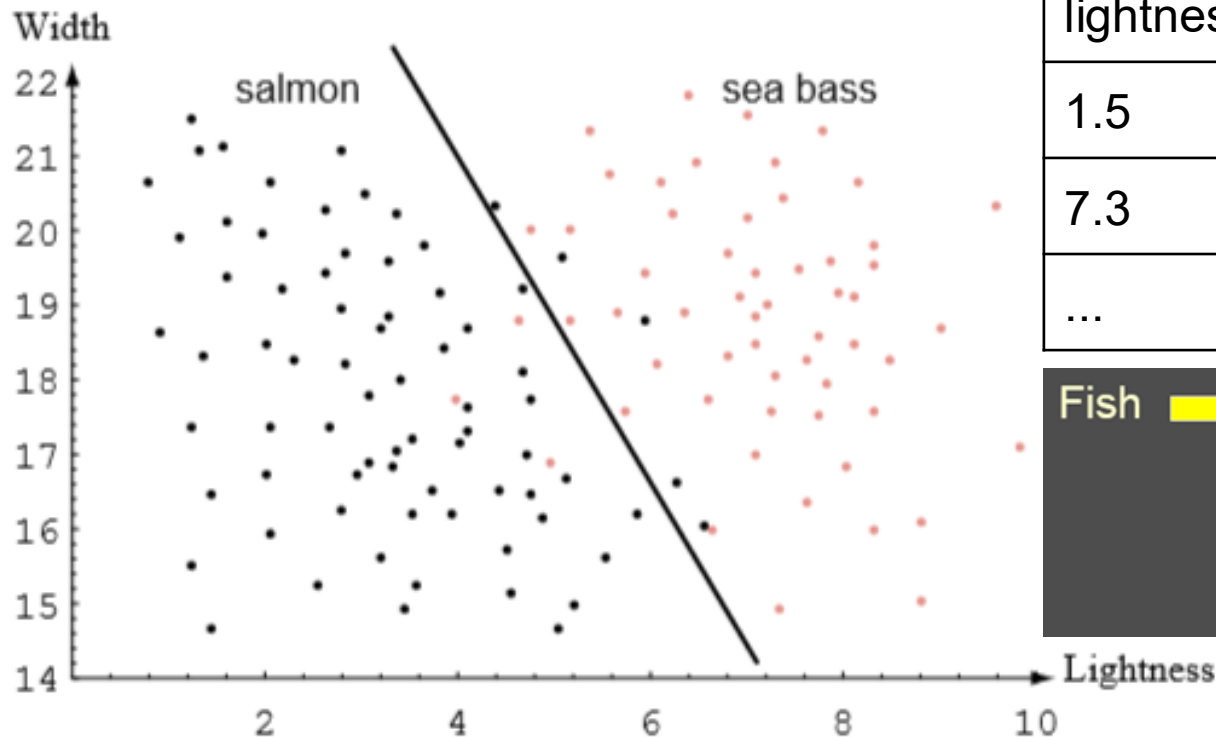
Misclassification



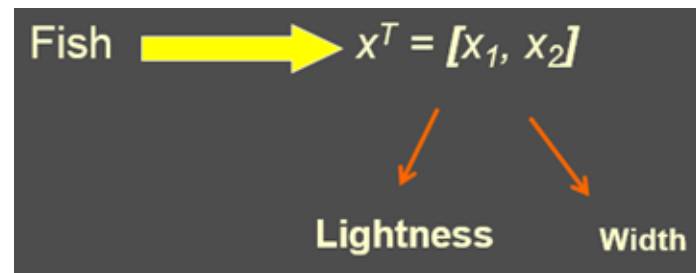
salmon in “sea bass” cans vs sea bass in “salmon” cans ?



Tentative Model: Lightness and Width Feature

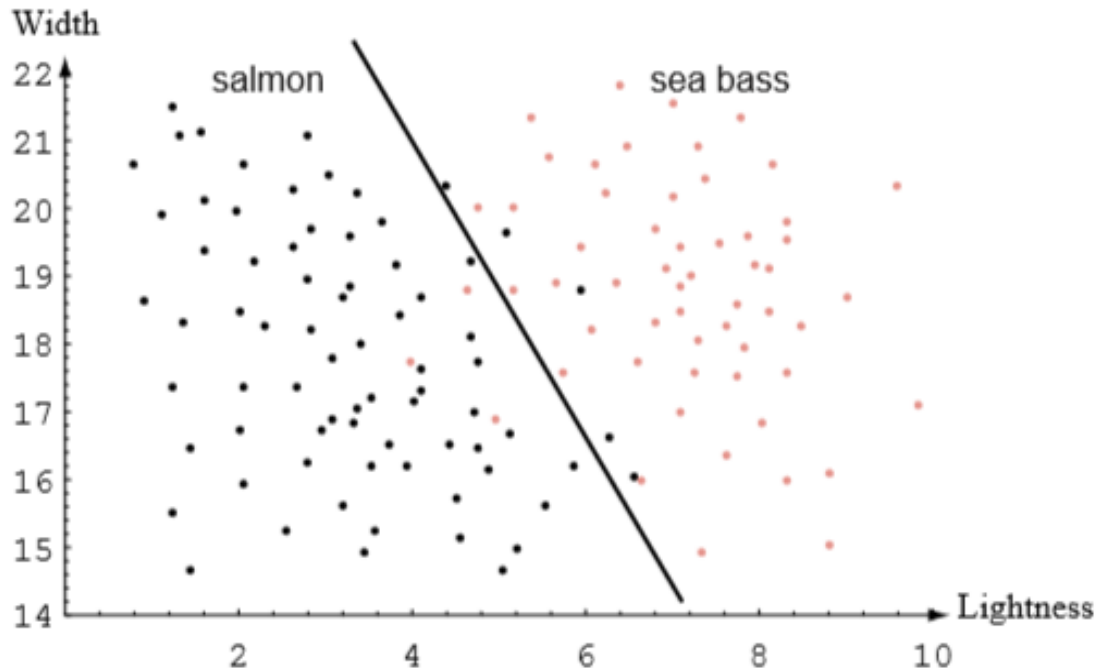


lightness	width	category
1.5	14.6	salmon
7.3	15	Sea bass
...		





Rule based on 2 Features

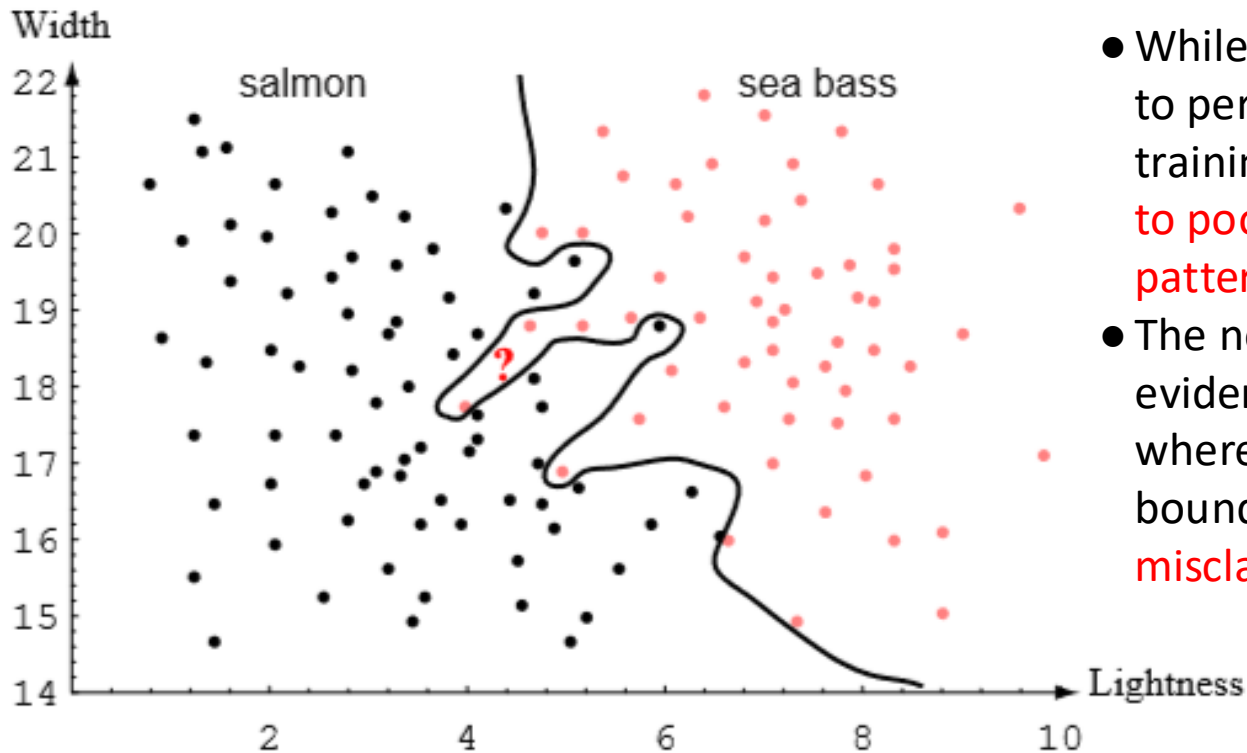


$$Y = a * \text{width} + b * \text{lightness} + c$$

Given decision boundary, the following rule for separating the fish: Classify the fish as sea bass if its feature vector falls above the decision boundary shown, and as salmon otherwise.



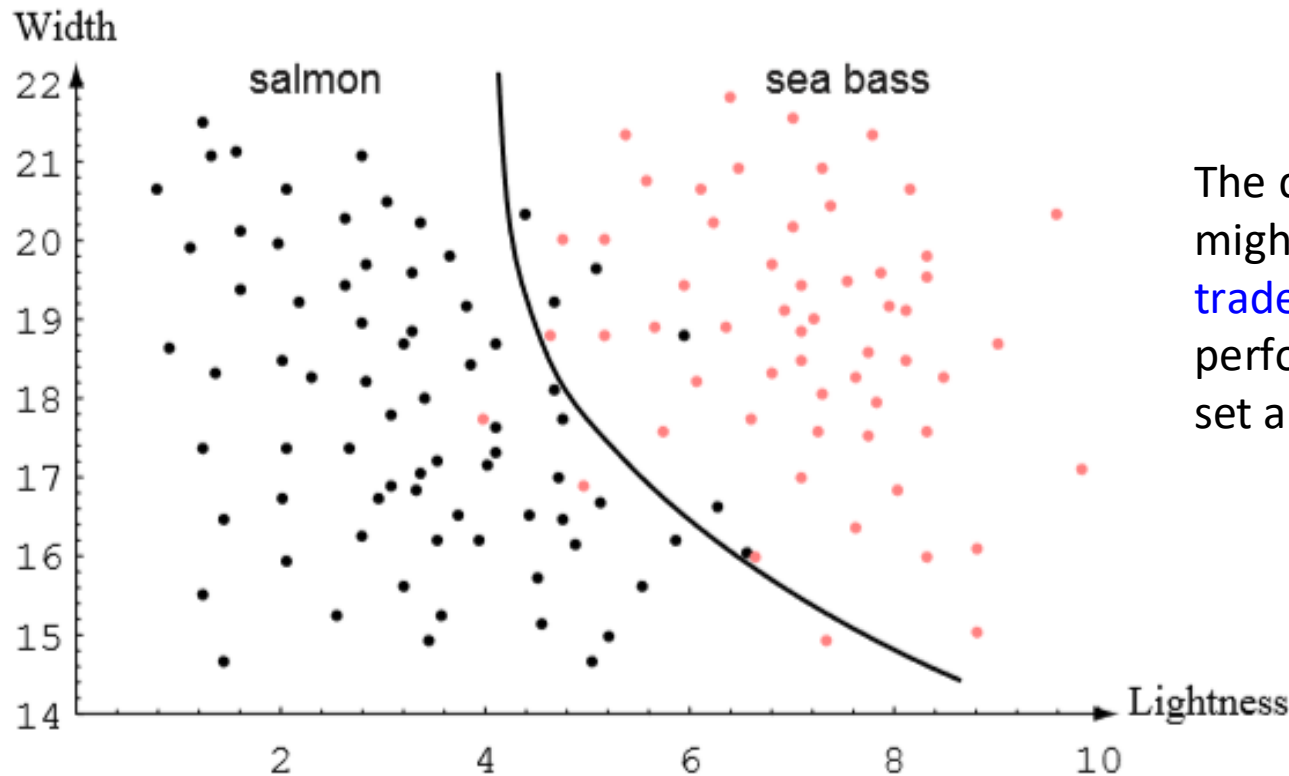
Tentative Complex Model: Training Accuracy 100%



- While such a decision may lead to perfect classification of our training samples, it would **lead to poor performance on future patterns**.
- The novel test point marked **?** is evidently most likely a salmon, whereas the complex decision boundary shown leads it to be **misclassified** as a sea bass.



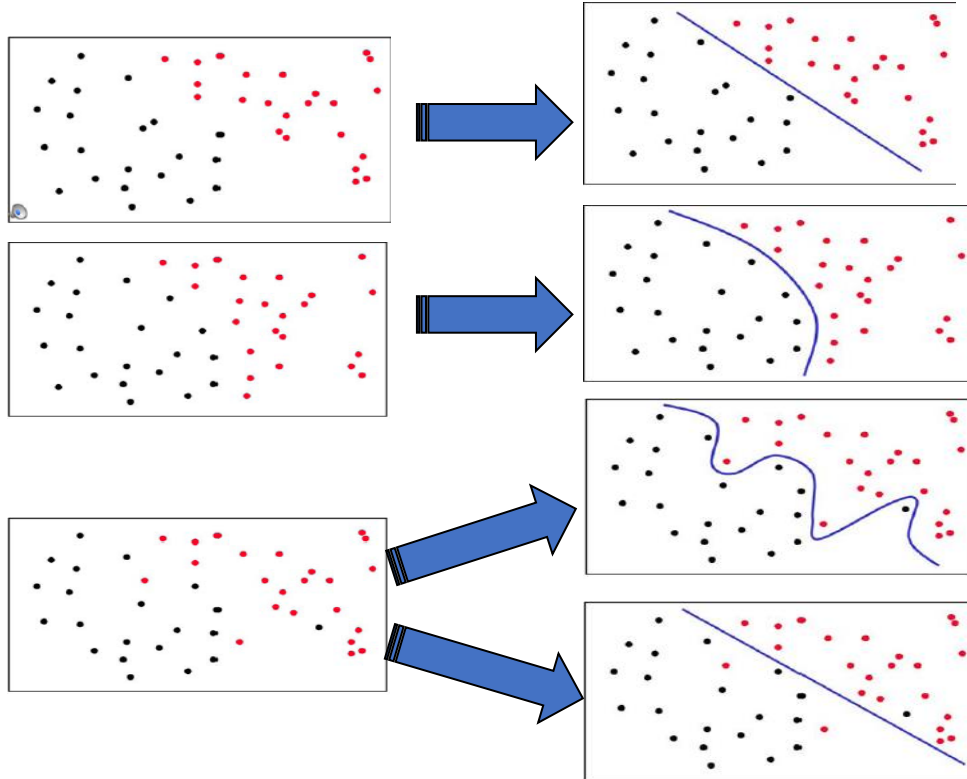
Optimal Model: Better Generalization



The decision boundary shown might represent the **optimal tradeoff** between performance on the training set and simplicity of classifier.

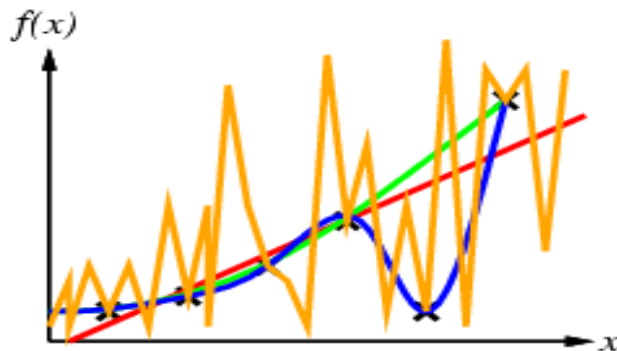
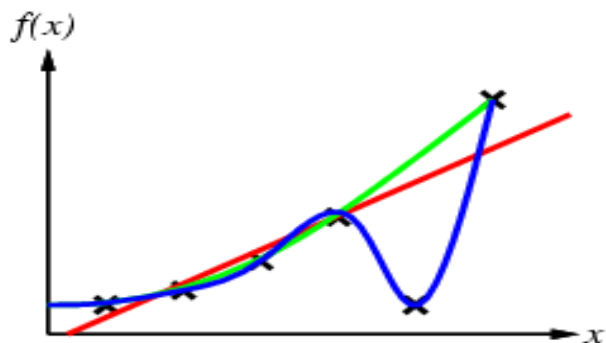
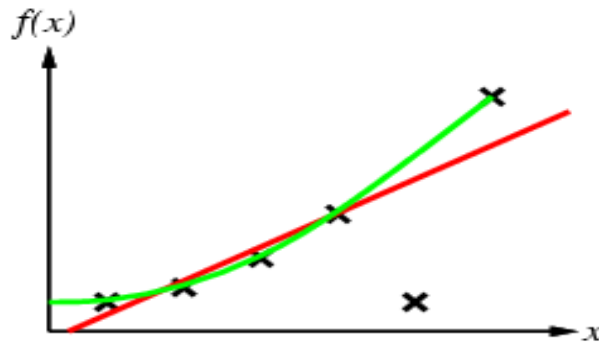
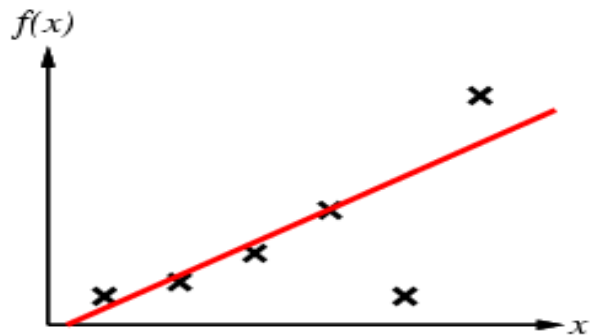
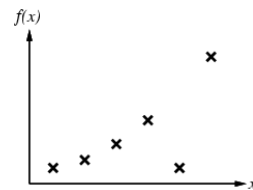


Curve Fitting: Example



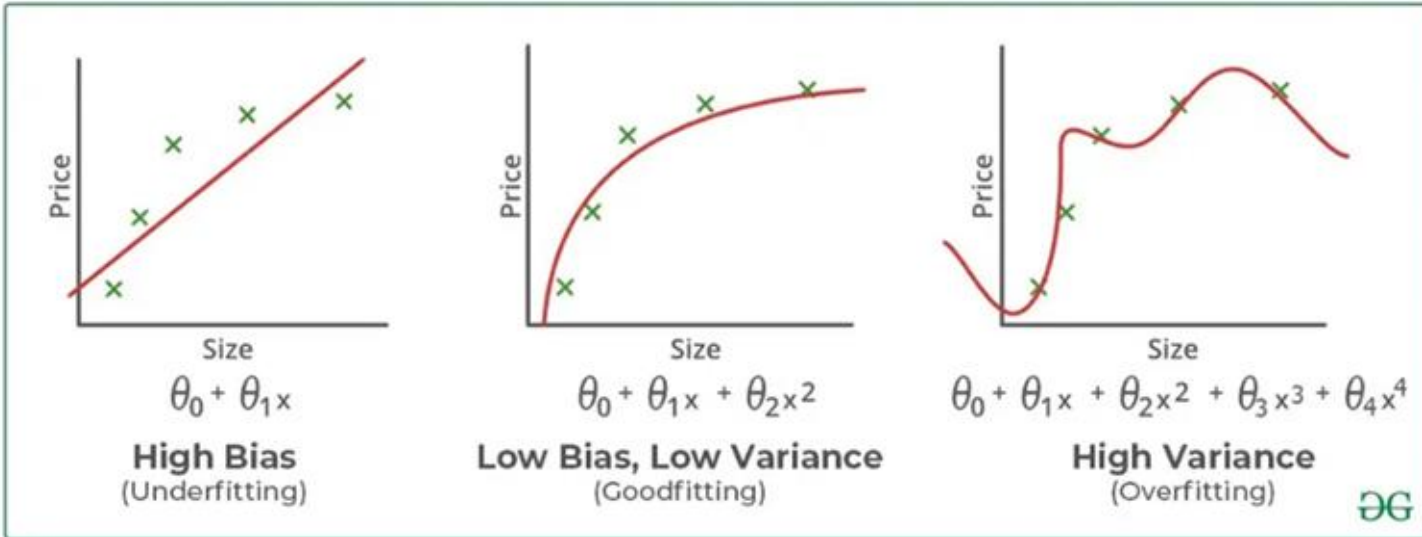


Curve Fitting: Example 2





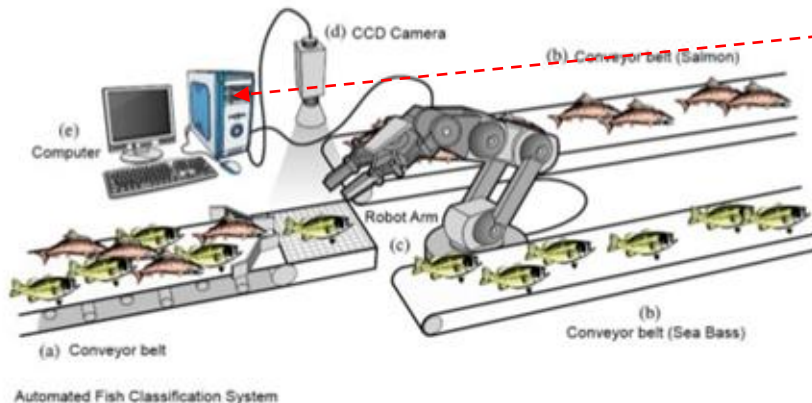
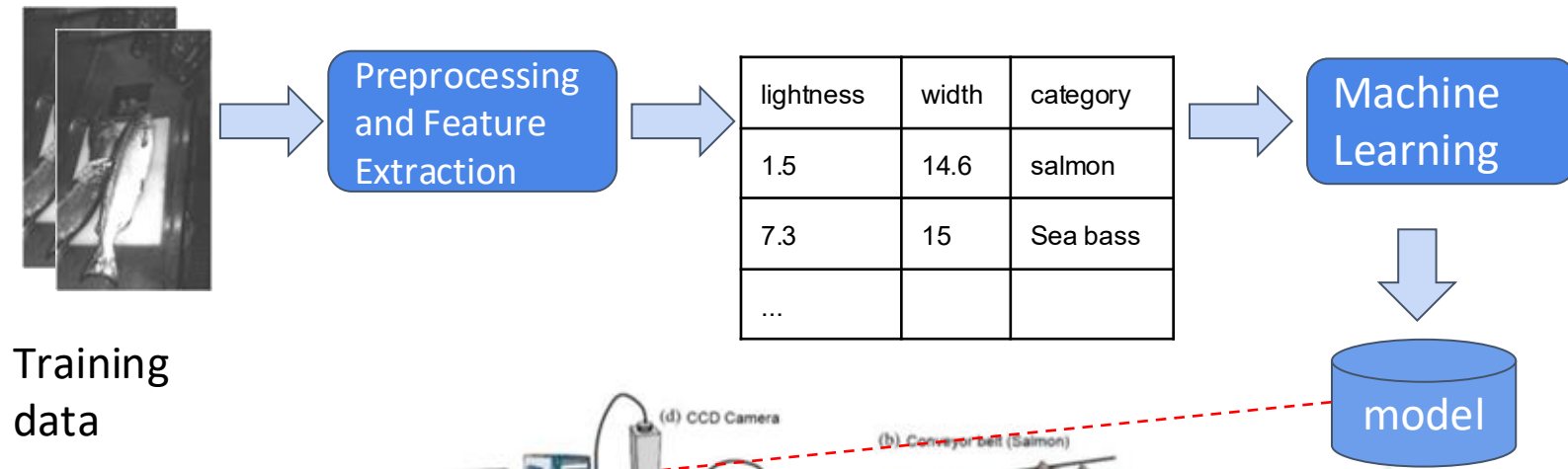
Bias vs Variance



Sumber gambar: <https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>



Modeling using Supervised Learning





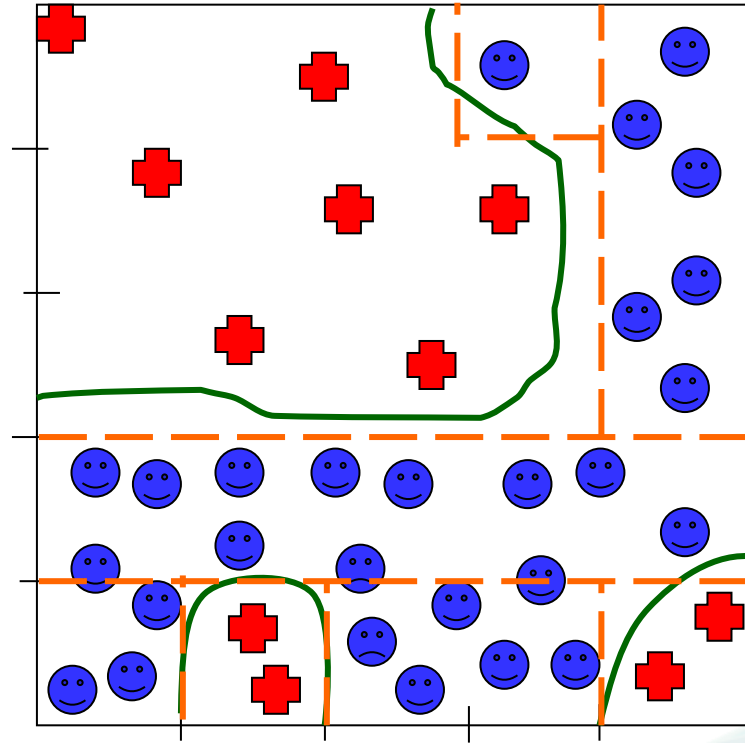
Data Set (Examples)

- Attribute-based representations
- Examples described by **attribute values**

Case	Age	Prescription	Astigmatism	Tear Production	Lens
1	young	myope	not astigmatic	reduced	none
2	young	myope	not astigmatic	normal	soft
3	young	myope	astigmatic	reduced	none
4	young	myope	astigmatic	normal	hard
5	young	hypermetrope	not astigmatic	reduced	none
6	young	hypermetrope	not astigmatic	normal	soft
7	young	hypermetrope	astigmatic	reduced	none
8	young	hypermetrope	astigmatic	normal	hard
9	pre-presbyopic	myope	not astigmatic	reduced	none
10	pre-presbyopic	myope	not astigmatic	normal	soft
11	pre-presbyopic	myope	astigmatic	reduced	none
12	pre-presbyopic	myope	astigmatic	normal	hard
13	pre-presbyopic	hypermetrope	not astigmatic	reduced	none
14	pre-presbyopic	hypermetrope	not astigmatic	normal	soft
15	pre-presbyopic	hypermetrope	astigmatic	reduced	none
16	pre-presbyopic	hypermetrope	astigmatic	normal	none
17	presbyopic	myope	not astigmatic	reduced	none
18	presbyopic	myope	not astigmatic	normal	none
19	presbyopic	myope	astigmatic	reduced	none
20	presbyopic	myope	astigmatic	normal	hard
21	presbyopic	hypermetrope	not astigmatic	reduced	none
22	presbyopic	hypermetrope	not astigmatic	normal	soft
23	presbyopic	hypermetrope	astigmatic	reduced	none
24	presbyopic	hypermetrope	astigmatic	normal	none



Data Mining Technique: Machine Learning



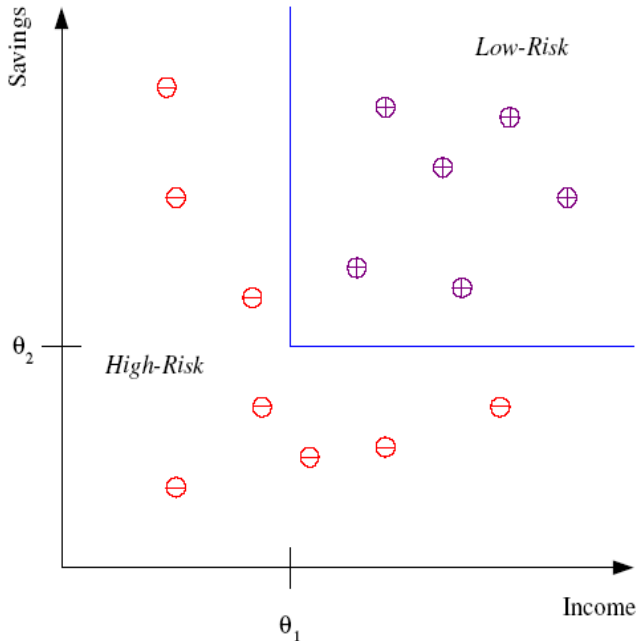
- Non-chumer
- chumer
- "Real" Function
- Learning Result

Learning: finding pattern
Pattern: Function Approximation



Example: Credit Scoring

- Differentiating between **low-risk** and **high-risk** customers from their *income* and *savings*



Discriminant: IF *income* $> \theta_1$ AND *savings* $> \theta_2$

THEN **low-risk** ELSE **high-risk**



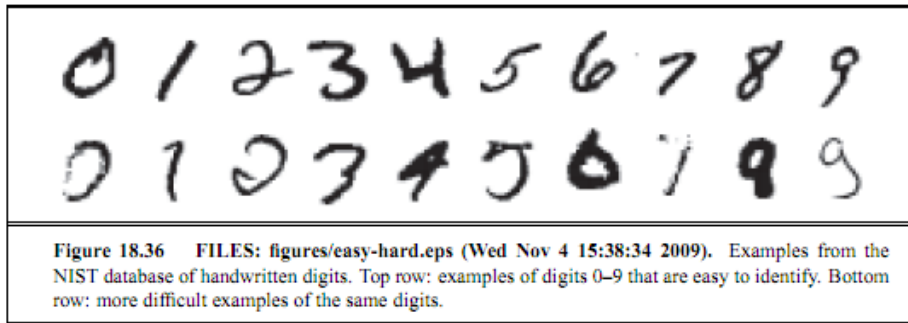
Applications for Supervised Learning

Given: Training examples $(\mathbf{x}; f(\mathbf{x}))$ for some unknown function f

Find: A good approximation to f .

Example Applications

- **Handwriting Recognition**
 - $\mathbf{x}$: Data from pen motion.
 - $f(\mathbf{x})$: Letter of the alphabet.





Applications for Supervised Learning

- **Disease diagnosis**
 - $\mathbf{x}$: Properties of patient (symptoms, lab tests)
 - $f(\mathbf{x})$: Disease (or maybe, recommended therapy)
- **Face recognition**
 - $\mathbf{x}$: Bitmap picture of person's face
 - $f(\mathbf{x})$: Name of the person.
- **Music genre classification**
 - $\mathbf{x}$: music file (musicXML, analog file, MIDI)
 - $F(\mathbf{x})$: music genre
- **Sentiment analysis**
 - $\mathbf{x}$: tweets / status facebook
 - $F(\mathbf{x})$: sentimen (positif/negatif)





Applications for Supervised Learning

- **Spam Detection**
 - $\mathbf{x}$: Email message
 - $f(\mathbf{x})$: Spam or not spam.
- **Situations where there is no human expert**
 - $\mathbf{x}$: Bond graph for a new molecule.
 - $f(\mathbf{x})$: Predicted binding strength to AIDS protease molecule.
- **Situations where humans can perform the task but can't describe how they do it.**
 - $\mathbf{x}$: Bitmap picture of hand-written character
 - $f(\mathbf{x})$: Ascii code of the character
- **Situations where the desired function is changing frequently**
 - $\mathbf{x}$: Description of stock prices and trades for last 10 days.
 - $f(\mathbf{x})$: Recommended stock transactions
- **Situations where each user needs a customized function f**
 - $\mathbf{x}$: Incoming email message.
 - $f(\mathbf{x})$: Importance score for presenting to user (or deleting without presenting).





Summary

Supervised
learning: inductive
learning, find
hypothesis

Classification vs
regression

Hypothesis to
predict unseen
data

Fish packing plant:
case study

Intro to Data Science



Thank You

